

Michealmas Week 5 & 6

1w703

1 Completed Task

I have came up with the method for multi-step DiKL to increase its capacity.
To formalize this approach, we optimize the following objective:

$$\theta = \arg \min_{\theta} \text{DiKL} \left(f_t^{\theta} \left(\tilde{f}_{t-1}(x) \right) * k_t \middle| \middle| p_d * \tilde{k}_t \right), \quad (1)$$

where $\tilde{k}_t$ represents the cumulative application of Gaussian convolution kernels:

$$\tilde{k}_t = k_T \circ k_{T-1} \circ \cdots \circ k_{t+1}. \quad (2)$$

This formulation allows the neural sampler to iteratively refine noisy samples, improving their quality over multiple steps.

Figure 1 illustrates this iterative denoising process. At each step, the function f_t removes a portion of the noise introduced by the forward process. The full sequence of transformations is expressed as:

$$\tilde{f}_{t-1} = f_{t-1} \circ f_{t-2} \circ \cdots \circ f_1. \quad (3)$$

The process starts from an initial noisy sample x_T and applies successive transformations to recover the clean sample x_0 .

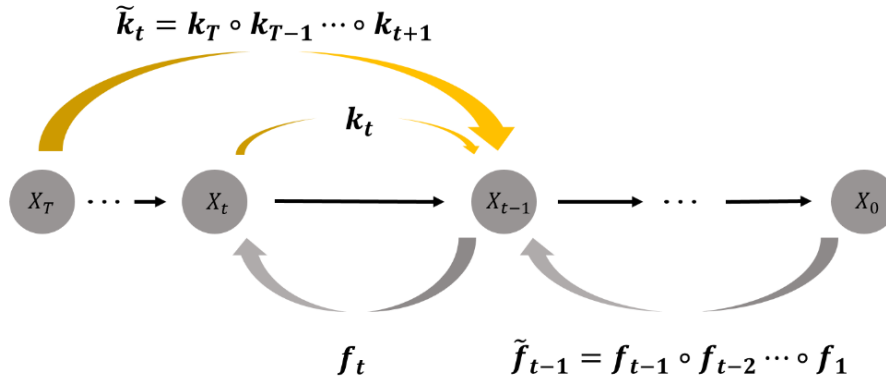


Figure 1: Illustration of the multi-step DiKL process, where the learned transformations progressively refine noisy samples.

For efficient training, the loss function derived in the report before is used so that we do not need to train a new score model for each step.